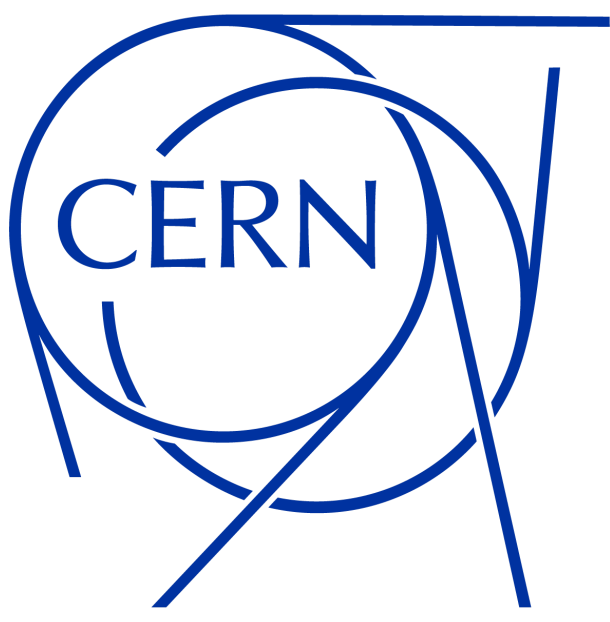




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CERN's PReservation As A Service connects live information systems to standardized OAIS-compliant preservation workflows. Through service-level agreements, data and metadata are harvested, from which Submission Information Packages (SIPs) are created, validated, enriched and transformed into Archival Information Packages (AIPs) for controlled, long-term preservation.

TECHNICAL

From live systems to trusted AIPs

Automated, customized pipelines designed for scale, repeatability, and resilience.

Preservation pipeline

The PReservation As A Service orchestrates a consistent workflow across diverse CERN source systems:

1 Harvest from source

2 Create standardized SIP

3 Validate SIP

4 Extract technical metadata

5 Create AIP

6 Notify source system

7 Push to cold storage

8 Update registry

Technology

The project is open source and built on open source software and dependencies.

Modular SIP-to-AIP processing

Current pipelines delegate preservation actions to Archivematica and its micro-services, including:

File identification

Anti-virus scanning

File normalization

Standardized METS

The concept of pluggable "SIP2AIP factories" is currently being explored by the EOSC EDEN project.

Engineering for CERN scale

- Scalable parallel processing architecture built on Celery.
- Using the CERN Tape Archive, designed for petabytes of High Energy Physics data.
- Registry updates and source notifications for traceable preservation status.
- Monitoring and alerting for proactive pipeline management.

Resilience is the next frontier

Ongoing development of PReservation As A Service enables automatic restarting of failed or superseded workflows with revised parameters. The service must track failed or stuck processes, non-conforming packages, problematic conversions and evolving normalization rules.

POLICY & STRATEGY

One preservation mandate across CERN

Controlled scope, clear custody, well-defined responsibilities for diverse institutional data.

Defining what must be preserved

CERN's strategy establishes a predefined scope of **must-be-preserved data**, primarily based on historical value and described through a laboratory-wide operational circular.

Institutional value

- Consistent OAIS standard across many data types.
- Well-defined responsibilities between source systems and the CERN Archives.
- Providing a centralized registry for CERN's AIPs.

Access control by design

The highest level of access restriction is applied to archived records until precise access conditions are defined by the archivists. Access rules from live information systems are not propagated, avoiding synchronization delays and reducing the risk of exposing restricted or sensitive records.

Three stewardship scenarios

1 **Central harvesting:** source systems are harvested by the preservation service which creates and maintains AIPs, and stores them in the CERN Tape Archive.

2 **Submitted SIPs:** source systems create SIPs and submits them to the service, which produces, stores, and maintains the resulting AIPs.

3 **Locally managed AIPs:** source systems create and manage their own AIPs while publishing their descriptions to the central registry.

Preservation in operation

CERN Institutional Repository
Records are harvested after a grace period; preservation status is displayed through a preservation label in the CERN Document Server.

International Linear Collider Document Server
Content was preserved before the legacy system was phased out and is now available on demand from the Digital Archives.

Mailboxes
The mailboxes of the Directorate are preserved and stored with a 30-year access embargo.

Expanding coverage

Target content types for upcoming repository integrations include multimedia, collaboration notes, source code, conference events, scientific datasets, and technical drawings.

PReservation As A Service in numbers - 2026

